9C Lesson 6

Ecology: abundance and distribution

Year 9 Science · 9C Threat from disease · Student Book pages 48 to 49

Name _____ Class _____ Date _____

Today I can:

- ☐ I can choose suitable apparatus for measuring the abundance and distribution of organisms.
- ☐ I can use data from a sample to estimate a population size and say how accurate the estimate is.

Key words 关键词

abundance	丰度
distribution	分布
population	种群
sample	样本
estimate	估算
quadrat	样方
transect	样线
pitfall trap	陷阱
sweep net	捕虫网
random	随机的
biodiversity	生物多样性

Command words 指令词

Calculate 计算 · Complete 完成（填写） · Describe 描述 · Draw 画出 · Estimate 估算 · Explain 解释 · Give 给出 · Match 连线（配对） · Predict 预测 · Suggest 提出（建议） · Tick 打勾

● Core Everyone does this section.

1. Match each word to its definition. Draw a line.

abundance	an approximate calculation of something
distribution	the number of different species in an area
sample	how common a species is in a habitat
estimate	a small selected part of the total number
biodiversity	how organisms are spread out in an area

2. Tick one box in each row to show the best method for sampling each organism.

Organism	pitfall trap	sweep net	quadrat	pooter
ground beetles				
insects in long grass				
daisies on a lawn				
tiny insects on a leaf				
limpets on a rocky shore				

3. Students counted one species of snail in eight quadrats placed at random in a field 30 m by 25 m. Each quadrat was 0.5 m by 0.5 m.

Quadrat	1	2	3	4	5	6	7	8
Number of snails	3	1	0	2	4	1	2	3

a) Calculate the total number of snails counted.

b) Calculate the total area sampled by the eight quadrats, in m2.

c) Calculate the area of the whole field, in m2.

d) Estimate the population of this snail species in the whole field. Show your working.

4. Record your own quadrat results, then estimate the population of your plant in the 10 m by 10 m area.

a) Complete the table.

Quadrat	Coordinates	Number of plants
1		
2		
3		
4		
5		
6		
7		
8		
total		

b) Estimate the population: total x 100 m2 / 2 m2. Show your working.

●● Stretch When Core is finished.

S1. Give one reason for using quadrats to estimate a population size rather than counting all the snails.

S2. Suggest the best way to decide where to place the eight quadrats in the field. Explain your answer.

S3. Snails of another species in the field gather in groups during the day. Explain how this could affect an estimate of their population size made with the same method.

●●● Challenge Explain in full sentences.

C1. Ecologists compared the biodiversity of birds in rainforest and in oil palm plantations that replaced some of the forest. They walked along transects through each area, recording every bird they saw or heard.

a) Suggest why the scientists used transects rather than quadrats.

b) Give two things the scientists should keep the same so that the results from the two areas can be compared fairly.

c) Oil palm plantations have trees planted in rows, all the same age; rainforest has many kinds of trees of different sizes. Predict in which area the birds would show a more regular distribution, and give a reason.

C2. There are many types of grass, each with slightly different leaf shapes. Describe how you would find out which of four species was the most abundant in a large field.

My thinking (any language 可用中文)

During an outbreak, scientists estimate the number of infected people from samples. Suggest one reason why the estimate might be too low, and one reason why it might be too high.

Word bank abundance 丰度 · distribution 分布 · sample 样本 · estimate 估算 · quadrat 样方 · transect 样线 · pitfall trap 陷阱 · sweep net 捕虫网 · random 随机的 · clumped · biodiversity 生物多样性

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Exit ticket 9C Lesson 6: Ecology: abundance and distribution Name _____

1. Why do ecologists take samples instead of counting every organism?

2. 5 daisies were found in one 0.25 m2 quadrat. The lawn is 40 m2. Estimate the daisy population.

3. Give one reason why quadrats should be placed at random.

Today I can: Goal 1 ☐ Goal 2 ☐